Development and Characterisation of Eggshell–Agar Biocomposites for Sustainable Interior Applications



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ICSMMEP-26 · Mecca, Saudi Arabia · 3–4 November 2026

State of the art I — eggshell valorisation

WHERE EGGSHELL WASTE CURRENTLY GOES

Application	Matrix	Reference
Cement and concrete replacement	Portland clinker	Teara et al. 2020
Laterite brick stabilisation	Cement	Adogla et al. 2016
Asphalt binder modification	Bitumen	Razzaq & Yousif 2017
Soil stabilisation	Lime substitute	Soundara et al. 2015
Ceramic wall tile	Ceramic body	Vilarinho et al. 2020
Particulate filler	PLA, epoxy, PVC, PU	Owuamanam & Cree 2020
Bio-packaging reinforcement	Starch, PLA films	Recent reviews, 2021–2025

Reviews: Ngayakamo & Onwualu, Heliyon 2022 · Owuamanam & Cree, J. Compos. Sci. 4(2):70, 2020 · Frontiers in Materials 9:1024977, 2022

THE PATTERN ACROSS THIS LITERATURE

Eggshell enters as a mineral filler. The binding phase it enters remains petrochemical, clinker-based or ceramic. Reported loadings sit between 5% and 30% of the matrix. The waste is valorised; the matrix is not.

WHY THE FIELD KEEPS RETURNING TO IT

- Approximately 94% calcium carbonate, chemically close to quarried limestone.
- A porous, layered shell microstructure with a protein-bound crystalline matrix.
- Specific gravity of 0.85–2.66, lighter than most mineral fillers.
- Continuously generated, geographically distributed, and currently landfilled at scale.

The material argument for eggshell is settled. The open question is what binds it.

State of the art II — agar as a structural binder

WHAT IS KNOWN ABOUT AGAR–GLYCEROL SYSTEMS

Hernández et al. (Materials 15:3954, 2022) mapped 32 agar–glycerin–water formulations by design of experiments, trained a regression model on the result, and released the dataset openly.

Across the agar film literature the direction is consistent: glycerol raises elongation and lowers tensile strength. Agar fraction and the agar–glycerol interaction, not glycerol alone, govern stiffness.

AND WHAT IS NOT

That body of work concerns unfilled thin films, tested in tension, for packaging. Agar has not been characterised as the binder of a mineral-filled cast solid in bending, which is the loading case for an interior surface component.

THE SHIFT THIS STUDY MAKES

—	Agar film literature	This study
	Thin cast film	Cast solid, 20–50 mm
	Unfilled or organic	Mineral, 66% of solids
	Uniaxial tension	Compression and bending
	Packaging	Interior surfaces
	Tensile, barrier	Strength, fire, hygric

The compositional logic is carried over from the film literature. The form, the filler, the loading case and the property set are not.

The gap is not rhetorical — it is documented

AN OPEN COMMONS ALREADY EXISTS

Materiom publishes biomaterial recipes under CC-BY-SA, excludes petrochemical ingredients, archives each recipe as prior art, and indexes property data on the Materials Data Facility with citable DOIs. The infrastructure for shared procedures is built.

BUT THE PROPERTY DATA IS THIN

Property data currently exists for four recipes:

- agar bioplastic
- gelatin bioplastic
- mycelium–walnut shell composite
- mussel shell–sucrose ceramic

4

recipes with published property data

> 50 000

articles queued for automated recipe extraction

WHAT THIS MEANS FOR THE PRESENT WORK

The nearest analogue in that set is the mussel shell–sucrose ceramic: a shell mineral in a biological binder. Nothing in it couples a shell mineral to an agar binder, and nothing reports compressive, flexural, ignitability and hygric data for the same formulation family.

So the contribution is not a claim that nobody has thought of this. It is a protocol and a property set entering a commons that is currently four recipes deep.

materiom.org/data, accessed October 2026

Aim, scope and research questions

AIM

To document a reproducible production protocol for a fully bio-based eggshell composite, and to characterise its mechanical, thermal and hygric response against both conventional and bio-based interior materials.

SCOPE

- Five formulations, two mechanical test programmes
- Three specimens per formulation
- Standardised tests: ASTM D790, ISO 11925-2, ASTM D570
- Single laboratory, ambient curing, no accelerated ageing

RESEARCH QUESTIONS

Q1 Which formulation variable governs compressive strength, and over what range?

Q2 What is the effect of plasticiser content on flexural behaviour?

Q3 How does the material compare with conventional interior surface materials?

Q4 How does it compare with other bio-based composites?

Q5 Is compressive strength the appropriate criterion for this material class?

Constituents and their function

Constituent	Origin	Function in the composite	Relevant property
Eggshell powder	Food waste	Mineral filler; load-bearing phase	94% $CaCO_3$; porous shell structure
Agar	Red seaweed	Binder; forms the continuous phase	Thermoreversible gel; high water retention
Glycerol	Vegetable oil	Plasticiser; controls ductility	Humectant; low vapour pressure
Water	—	Carrier; governs solids concentration	Fully evaporated during drying
Shellac	Kerria lacca resin	Surface finish; hydrophobic barrier	Biodegradable; film-forming

Design logic: the mineral phase carries load, the agar network holds it together, and glycerol sets where the composite sits on the brittle–ductile axis. Water is not a component of the final material; it is the process variable that determines how densely the solids pack.

Gelatine, starch and vinegar were trialled during development and are not part of the reported formulations.

Protocol I — preparation of the mineral fraction

1 **Collection and cleaning**

Shells from a single local bakery, collected over more than one year. Membrane and organic residue removed by hand.

2 **Thermal treatment**

Laboratory oven, 120 °C for 20 minutes. Described as decontamination and drying; sterility was not verified and dry-heat sterilisation requires higher temperatures.

3 **Cooling and drying**

Cooled to ambient, then dried until no residual moisture remained by mass.

4 **Milling**

Laboratory mechanical grinder; milling continued until a visually homogeneous fine powder was obtained.

5 **Sieving**

150 µm mesh. Only the passing fraction was used, consistent with eggshell powder procedures reported for polymer composites.



CONTROLLED PARAMETERS

Treatment temperature	120 °C
Treatment duration	20 min
Sieve aperture	150 µm
Collection period	> 12 months

Protocol II — the binder phase and its thermal regime

1 **Pre-mixing**

Water and glycerol combined and stirred to a homogeneous liquid before any solids are introduced.

2 **Agar dispersion**

Powdered agar added incrementally under continuous stirring. Bulk addition causes agglomeration that does not redisperse.

3 **Dissolution stage**

Heated to 95 °C and held until the agar is fully in solution. Below full dissolution the gel network does not form continuously.

4 **Cooking stage**

Reduced to 80 °C and held, stirring maintained, while viscosity develops into a pourable but cohesive binder.

THERMAL REGIME

95 °C

dissolution set point

80 °C

controlled cooking set point

55 min total heating, across both stages

The two-stage regime follows the 70–90 °C band reported for agar-based biomaterial preparation.

Reproducibility limit: stirring was manual throughout, so shear rate and mixing energy are not controlled quantities. The regime is reproducible in a studio; it is not yet specified tightly enough for an instrumented laboratory.

Protocol III — compositing, casting and drying

1

Mineral incorporation

Sieved shell fraction added incrementally into the binder held at 80 °C, under continuous stirring. Earlier addition causes sedimentation of the mineral phase.

2

Casting

Poured into silicone moulds and levelled. Silicone was selected for flexibility: it releases without a parting agent and without damaging the partially consolidated specimen.

3

In-mould setting

Four days. Mould geometry follows the test: 50 mm cube moulds for compression, flat moulds for flexural, flame and absorption specimens.

4

Ambient consolidation

Demoulded, then a further seven days at approximately 23 °C, shielded from direct sunlight and forced air movement. No oven drying at any stage.

DISCLOSURE — DIMENSIONAL LOSS

The composite shrinks appreciably as the water phase evaporates. Cross-sections were measured at the time of testing, but post-drying dimensions were not logged in the original study. Dimensions quoted in this presentation are mould dimensions unless stated otherwise.

Anyone applying this protocol should record specimen geometry after drying, not at casting.

23 °C · 11 days · no oven

Total time from casting to a finished specimen: 11 days, entirely at ambient temperature. No energy input beyond the 55-minute cook.

Formulation matrix and derived ratios

ID	Test	Water (g)	Shell (g)	Agar (g)	Glycerol (g)	Solids (g)	Shell : binder	Solids : water	Glycerol % of solids
C1	Compression	325	80	35	5	120	**2.00**	0.37	4.2
C2	Compression	310	90	40	5	135	**2.00**	0.44	3.7
C3	Compression	280	110	45	10	165	**2.00**	0.59	6.1
F1	Flexure	450	190	70	60	320	1.46	0.71	18.8
F2	Flexure	480	190	70	30	290	1.90	0.60	10.3

COMPRESSION SERIES

The shell-to-binder mass ratio is held at exactly 2.00 across all three formulations. The single variable that changes is solids concentration, which rises from 0.37 to 0.59.

FLEXURAL SERIES — CONFOUND

Glycerol was halved, but total batch mass was held at 770 g, so water rose to compensate. Shell-to-binder and solids concentration therefore moved as well. The glycerol effect is not fully isolated.

Test programme

Property	Standard	Specimen geometry	n	Measured quantity
Compressive behaviour	— (cube compression)	50 mm cube moulds	3 per mix	Maximum load; compressive stress
Flexural behaviour	ASTM D790, three-point	180 × 60 × 20 mm, 120 mm span	3 per mix	Flexural strength; flexural modulus
Reaction to fire	ISO 11925-2	40 × 50 mm, 45°, 15 s edge flame	2 ignitions	Time to ignition; time to self-extinguish
Water absorption	ASTM D570	40 × 50 mm, shellac-coated	1	Mass gain after 24 h immersion
Bulk density	Gravimetric	50 mm cube	1	Mass / mould volume

REPORTING CONVENTION

All mechanical values reported in this presentation are means of three specimens. Individual specimen values were not retained in the original study, so standard deviations cannot be reported. Fire, absorption and density figures rest on single determinations and are presented as indicative rather than characteristic values.

All testing: Advanced Construction Materials Research Laboratory, Hacettepe University.

Compressive strength scales with solids concentration

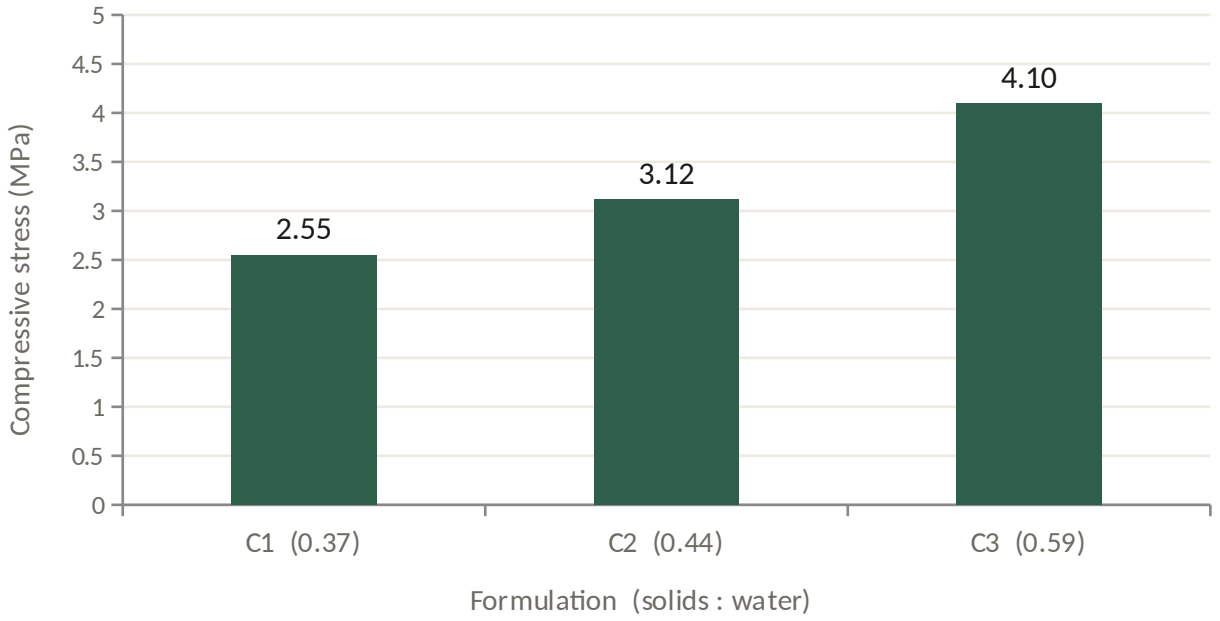
ID	Solids : water	F_{max} (kN)	σc (MPa)
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	6.80	4.10

OBSERVED RELATIONSHIP

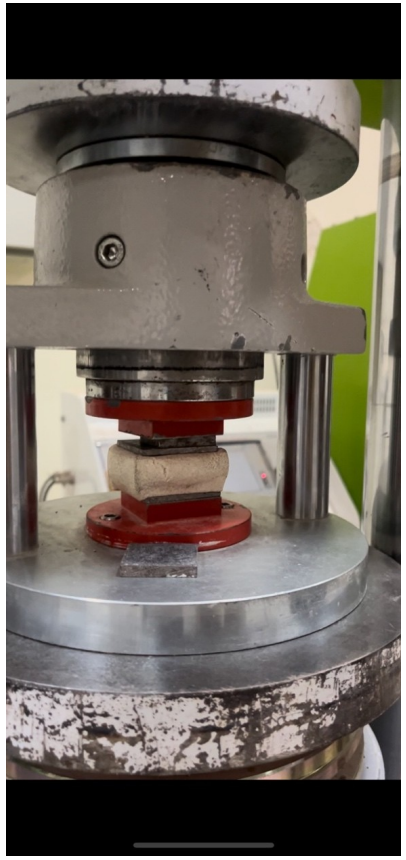
$$\sigma c = 6.93 \cdot (\text{solids} : \text{water}) + 0.04$$

$R^2 = 0.994$ over the range 0.37 – 0.59

With shell : binder fixed at 2.00, compressive stress is a linear function of solids concentration across the range tested. Three formulations cannot establish this as a law — but it is a testable hypothesis, and it predicts that strength is set by packing density rather than by mineral content as such.

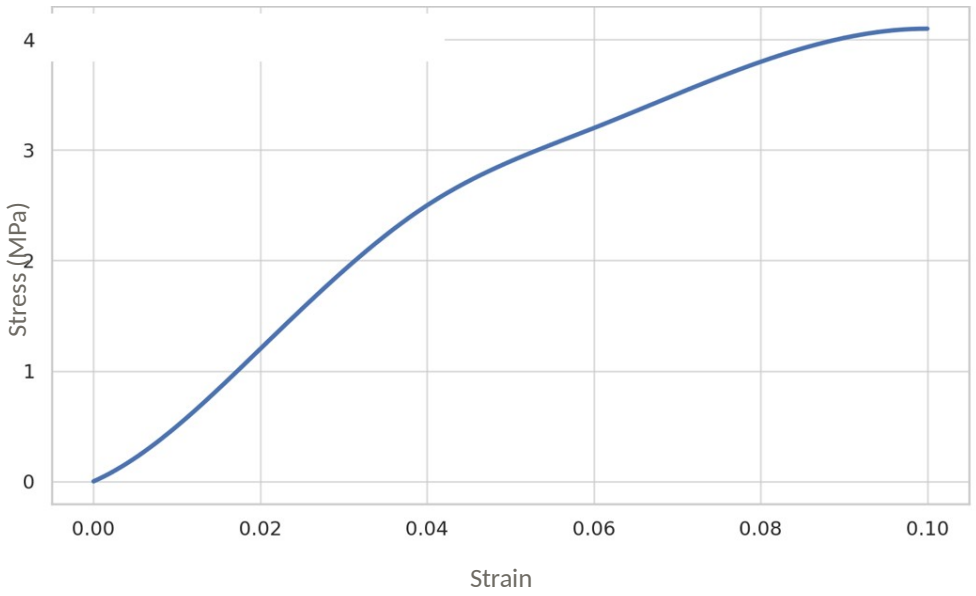


Failure mode: progressive crushing, not fracture



C3 specimen after loading: barrelled, shortened, still a single body

Compressive stress – strain, C3



Bulk density 1.38 g/cm³ (mould volume basis) · specific compressive strength 2.96 MPa per g/cm³

READING THE CURVE

- Initial linear elastic region to approximately 1 MPa.
- Progressive stiffness loss as the agar network yields and the mineral phase redistributes.
- Peak stress reached at roughly 10% strain, with no abrupt load drop.
- Beyond peak the specimen continues to deform while retaining integrity: energy is absorbed rather than released.

Strain values were derived from an assumed 10% maximum deformation based on observation during testing, not from instrumented strain measurement.

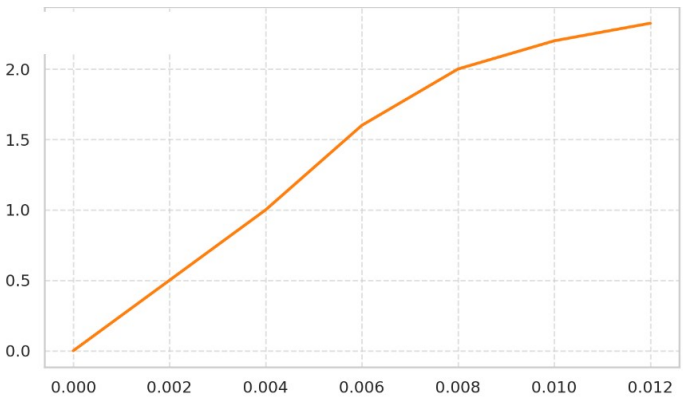
Flexural behaviour: plasticiser content is decisive

ID	Glycerol % of solids	F_{max} (N)	σ_F (MPa)	E_F (MPa)	ε at peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	**2.325**	**192**	0.0121

$\sigma_F = 3FL / 2bd^2$ with F = 310 N, L = 120 mm, b = 60 mm, d = 20 mm · $\varepsilon = \sigma_F / E_F$ (Hooke, elastic region)







Three-point rig at high deflection; kerfed panel arched by hand

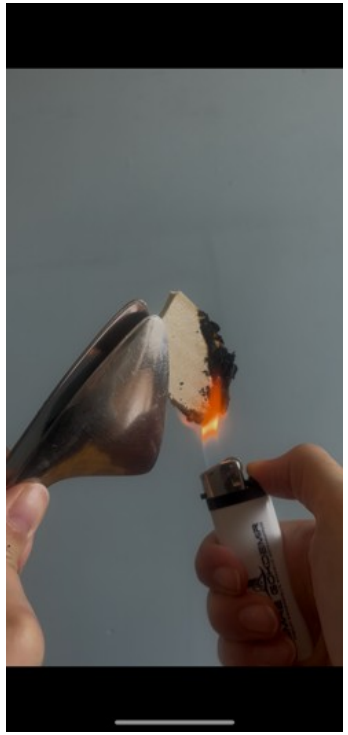
DISCLOSURE — SECTION GEOMETRY

The specimen carries transverse kerfs on its tension face, visible in both photographs. ASTM D790 assumes a solid rectangular section and the calculation above uses the full 60 × 20 mm. Part of the measured compliance is geometric, not material, and the reported modulus is therefore a lower bound on the bulk material.

INTERPRETATION

At 18.8% glycerol the composite deformed without developing measurable resistance — the testing machine registered no usable load. Halving the plasticiser produced a section that resists while still flexing. The transition is sharp, and it sits between these two values.

Reaction to fire, hygric response and density



IGNITABILITY — ISO 11925-2

40 × 50 mm dried specimen; flame at 45° to the upper corner for 15 s, repeated at the lower corner.

~10 s
to ignition

4 s
to self-extinguish

- Charring local to the flame contact point.
- No lateral flame spread in either ignition.
- No melting, no dripping — no thermoplastic behaviour.

Probable mechanism: endothermic carbonate decomposition releasing CO_2, diluting the flame front locally. Not verified calorimetrically. ISO 11925-2 is an ignitability test — no Euroclass classification, cone calorimetry, smoke or toxicity data were obtained.

WATER ABSORPTION — ASTM D570

40 × 50 mm specimen, three shellac coats, 10 g initial mass, 24 h full immersion.

0.7%

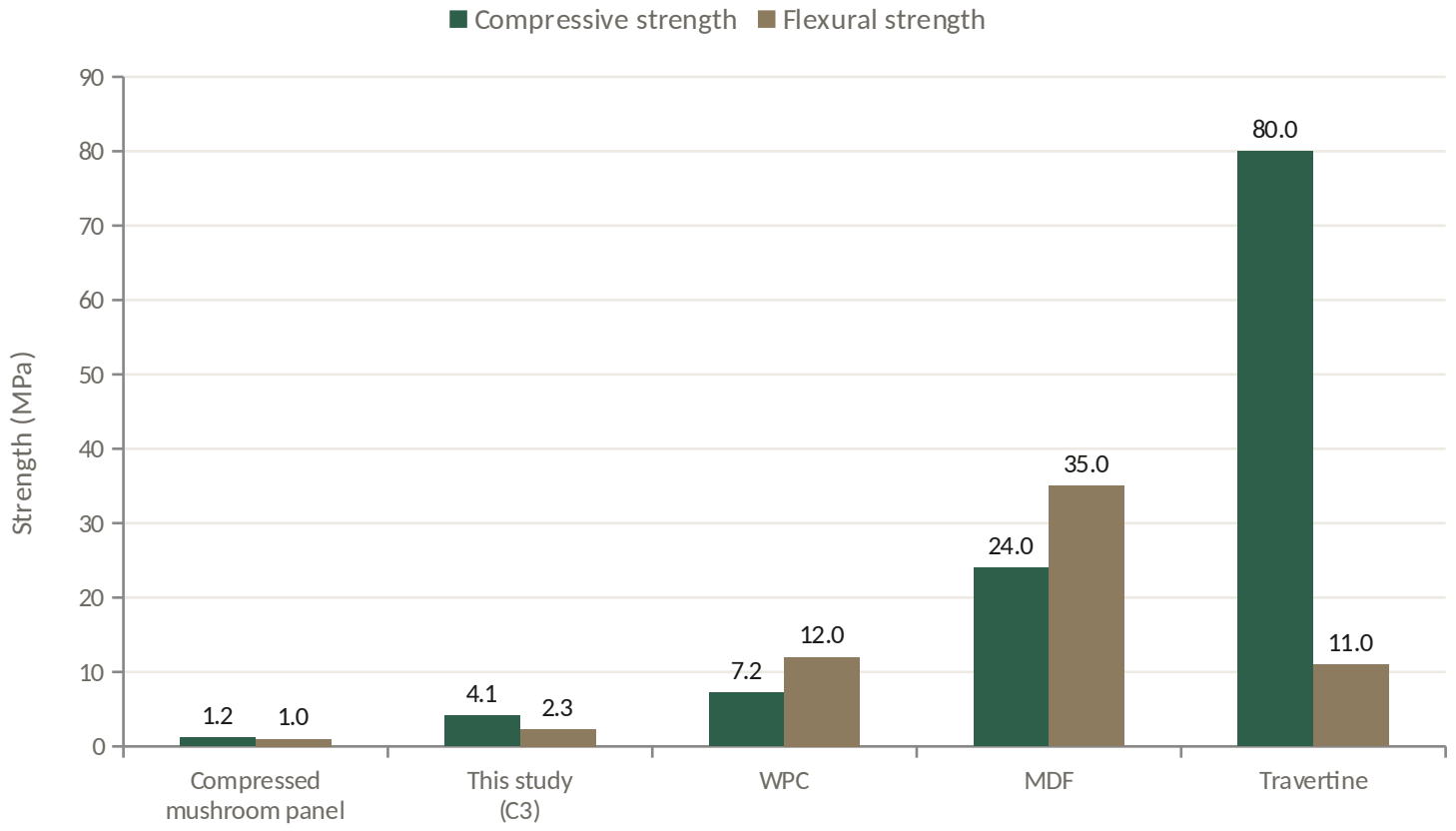
mass gain over 24 h. Slight dimensional expansion; local dissolution where the coating was thin. Coating integrity, not the composite, is the limiting factor for humid use.

BULK DENSITY — COMPARATIVE POSITION

Material	g/cm³	Reference
This study (C3)	**1.38**	—
Mycelium, cellulose composites	0.3 – 0.8	Jones et al. 2020
Fired clay brick	1.6 – 2.0	William & Owen 1995
Normal-weight concrete	2.4	William & Owen 1995

Quoted on a mould-volume basis and therefore a lower bound.

Benchmark I — conventional interior surface materials



Velasco & Gaona 2011 · Klyosov 2007 · Nemli & Aydın 2007 · Altun 2006

POSITION

Compressive strength is one sixth that of MDF and one twentieth that of travertine — a stone the material visually resembles.

It exceeds compressed mushroom panel, the only other fully bio-based board in this set, by a factor of three.

On strength alone, the material does not compete with engineered wood or stone, and this presentation does not claim that it does.

Benchmark II — bio-based composites

COMPRESSIVE STRENGTH

Material	σc (MPa)	Reference
This study (C3)	**4.1**	—
Mycelium-based composite	0.145 – 0.452	Jones et al. 2020
Hempcrete	0.3 – 0.6	Arnaud & Gourlay 2012
Eggshell-filled PLA	2 – 5	Kim & Pal 2010
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. 2016
30% algae-filled composite	21.3	Kim & Pal 2010

ELASTIC MODULUS

Material	E (MPa)	Reference
This study (F2)	**192**	—
Mycelium-based composite	1.8 – 5	Islam et al. 2018
Soy-based bioplastic	124	Mansour et al. 2021
Corn starch + hemp fibre	158	Dweib et al. 2004
Flax-fibre PLA	198	Faruk et al. 2012
LDPE	100 – 300	Zhao et al. 2022
[illegible]	[illegible]	Pope & Hudson

WHERE THE MATERIAL SITS

An order of magnitude above the fungal and lime-binder family in compression, and squarely inside the bio-polymer band in stiffness — between soy bioplastic at 124 MPa and flax-fibre PLA at 198 MPa, within the LDPE range. **A mineral-filled, water-cast, oven-free material reaches the stiffness of melt-processed bio-polymers. Unfilled agar films reported in the packaging literature sit one to two orders of magnitude below this; the mineral phase, not the binder, is carrying the stiffness.**

A derived property: modulus of resilience

DEFINITION

$$U_r = \sigma_F^2 / 2E_F$$

The elastic strain energy absorbed per unit volume up to the onset of permanent deformation — the area under the stress–strain curve to yield.

$$U_r = 2.325^2 / (2 \times 192) = 0.0141\ \text{MPa} = 14.1\ \text{kJ/m}^3$$

from this study's own measured σ_F and E_F for F2

Stiffness sits in the denominator. The low modulus reported as a deficiency in Benchmark I is precisely what raises this quantity.

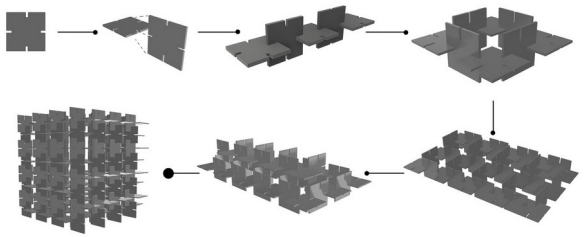
WHY IT FITS THIS MATERIAL CLASS

Both test programmes produced the same qualitative result: the specimens deformed substantially and retained integrity rather than fracturing. Strength figures record the load at which a material stops performing. They do not record whether it then fails gracefully or catastrophically — which, for a demountable non-loadbearing component, is the behaviour that governs service life.

PROPOSAL, AND ITS CURRENT LIMIT

Report σ_F and E_F from the same specimen set for bio-based materials, so that U_r can be derived. At present it rarely can: the literature reports strength and modulus in separate studies on different formulations, so a comparative U_r table cannot yet be assembled. This value is offered as a single data point and a measurement convention, not as a ranking.

Product proposal — a modular, demountable interior system



DERIVED FROM THE MEASURED PROPERTIES, NOT IMPOSED ON THEM

Low strength, high deformation capacity → component carries its own weight and handling loads only; no structural duty.
Self-extinguishing, no dripping → acceptable for interior surfaces away from ignition sources.
Coating-dependent water resistance → dry interior environments; not wet rooms or facades.
Repeatable small-format casting → a module, not a sheet. Failure is replaced one unit at a time.



SYSTEM CHARACTERISTICS

Reversible assembly — repeated cast units joined without adhesive, so components can be recovered intact.
Distributed load path — units interlock and share load across neighbours rather than transferring it through a frame.
Unit replaceability — a damaged unit is exchanged without dismantling the assembly; no single component is critical.
Formulation-family variation — colour, texture and format vary from pigments and mould geometry at constant composition.

Potentials for sustainable interiors

Waste valorisation at source

A locally available food-waste stream becomes a surface material without an intermediate industry. The shells for this study came from one bakery.

Low-energy production

No oven, no autoclave, no extrusion. A single 55-minute cook near 95 °C, then ambient drying. The thermal budget is a fraction of ceramic or cement-based surfaces.

End-of-life by design

Every constituent is biodegradable, including the shellac finish. The material returns to a biological cycle rather than requiring separation and disposal.

Reversible assembly

Adhesive-free modular assembly means the material can be recovered intact, so reuse precedes recycling in the hierarchy.

Replacing a mineral extraction

The calcium carbonate here substitutes quarried limestone in a surface application, which is a direct displacement rather than a downcycled use.

Local, low-barrier manufacture

The protocol requires a grinder, a sieve, a hotplate and silicone moulds. Production can be distributed rather than centralised.

No life-cycle assessment was carried out. These are qualitative potentials grounded in the production protocol, not quantified environmental claims.

Limitations of the study

Dispersion not reported

Means of three specimens per formulation; individual values were not retained, so no standard deviations or coefficients of variation can be given.

Formulation confound in flexure

Glycerol, solids concentration and shell-to-binder ratio moved together between F1 and F2. The plasticiser effect is not isolated.

Section geometry in flexure

The ASTM D790 calculation treats a kerfed specimen as a solid rectangle. The reported modulus is a lower bound.

Strain not instrumented

Compressive strain was derived from an assumed 10% maximum deformation based on observation, not measured directly.

Shrinkage unquantified

Post-drying dimensions were not logged. Density is quoted on a mould-volume basis and is a lower bound.

Process not instrumented

Manual stirring leaves shear rate and mixing energy uncontrolled. Single laboratory, single operator, no inter-laboratory reproduction.

No durability data

No humidity cycling, thermal cycling, UV exposure, microbial resistance or accelerated ageing was performed.

No fire classification or LCA

Ignitability only, with no Euroclass determination, and no life-cycle assessment to support the environmental argument.

A protocol offered for others to repeat has to state where it is thin. Each of these is addressable, and each defines a specific next experiment.

Conclusions

01 **A reproducible protocol**

Every parameter required to produce the composite is reported: 120 °C shell treatment, 150 µm fraction, 95 °C dissolution, 80 °C cook, 55 min, silicone casting, 11 days ambient.

02 **Solids concentration governs strength**

With shell : binder fixed at 2.00, compressive stress is linear in solids concentration across 0.37–0.59 (R^2 = 0.994, three formulations). Strength is set by packing density.

03 **Plasticiser sets the brittle–ductile transition**

Between 18.8% and 10.3% glycerol the composite moves from unmeasurable compliance to 2.33 MPa at 192 MPa.

04 **Bio-polymer stiffness, oven-free**

192 MPa places a water-cast, ambient-dried material inside the band occupied by melt-processed bio-polymers.

05 **Strength is the wrong sole criterion**

Deformation capacity governs service behaviour for demountable non-loadbearing components. σ and E should be reported together so that Ur can be derived.

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

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Standards: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018). Full reference list available on request.